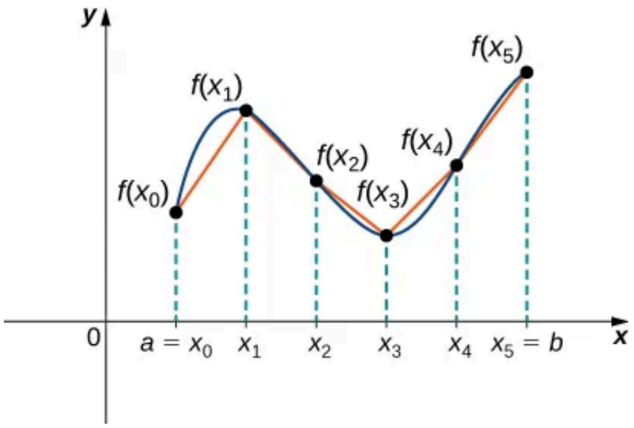
1 Arc Length of a Curve

Intuitively, the *arc length* of a curve is the distance one would travel if they were walking along the curve from start to finish. Here, we use a definite integral to calculate the arc length of a curve which is the graph of a function $f(x)$ defined on $[a,b]$. **We further assume that $f'(x)$ exists and is continuous on $[a,b]$.**

We start by using line segments to approximate the length of the curve, as follows. Let $P = \{x_i\}$ be a partition of $[a,b]$ into n subintervals of equal length Δx. For each $i = 1,\ldots,n$, construct the line segment from the point $(x_{i-1}, f(x_{i-1}))$ to the point $(x_i, f(x_i))$. This is illustrated for $n = 5$ in the figure below:



The length of each segment is

$$\begin{aligned}\ell_i &= \sqrt{(x_i - x_{i-1})^2 + (f(x_i) - f(x_{i-1}))^2} \\ &= \sqrt{(\Delta x)^2 + (\Delta y_i)^2} \\ &= \Delta x \sqrt{1 + \left(\frac{\Delta y_i}{\Delta x}\right)^2}\end{aligned}$$

where $\Delta y_i := f(x_i) - f(x_{i-1})$. Since f is differentiable on (x_{i-1}, x_i) and continuous on $[x_{i-1}, x_i]$, by the Mean Value Theorem there exists $x_i^* \in [x_{i-1}, x_i]$ such that $f'(x_i^*) = \frac{\Delta y_i}{\Delta x}$ and therefore

$$\ell_i = \sqrt{1 + (f'(x_i^*))^2}\Delta x.$$

The total arc length s is then approximately

$$s \approx \sum_{i=1}^{n} \sqrt{1 + (f'(x_i^*))^2}\Delta x.$$

Since f' is continuous, the limit as $n \to \infty$ exists, and we define the exact arc length to be

$$\boxed{s := \lim_{n\to\infty} \sum_{i=1}^{n} \sqrt{1 + (f'(x_i^*))^2}\Delta x = \int_a^b \sqrt{1 + (f'(x))^2}\,dx}. \tag{1}$$

1. Find the length of the graph of $y = 5x$ over the interval $[0, 2]$.

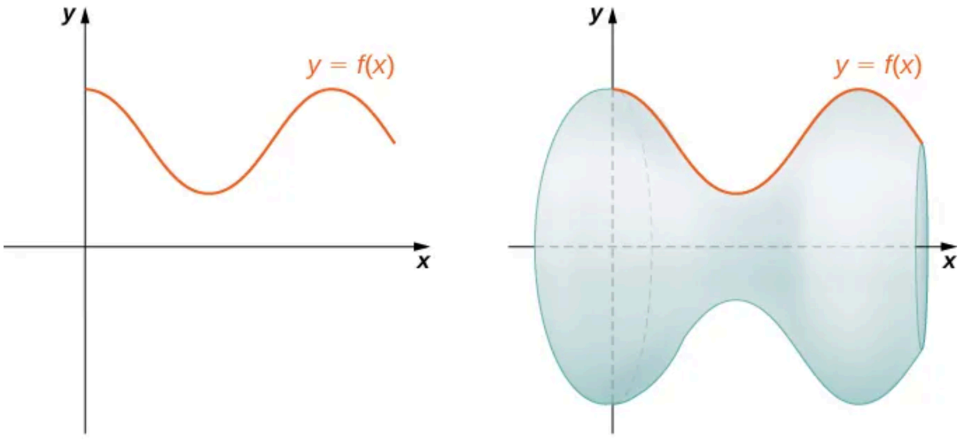
2. Let $f(x) = 2x^{3/2}$. Calculate the arc length of the graph of $f(x)$ over the interval $[0, 1]$.

The formula in Equation (1) often generates expressions that are difficult to integrate, and in practice one must approximate the integral.

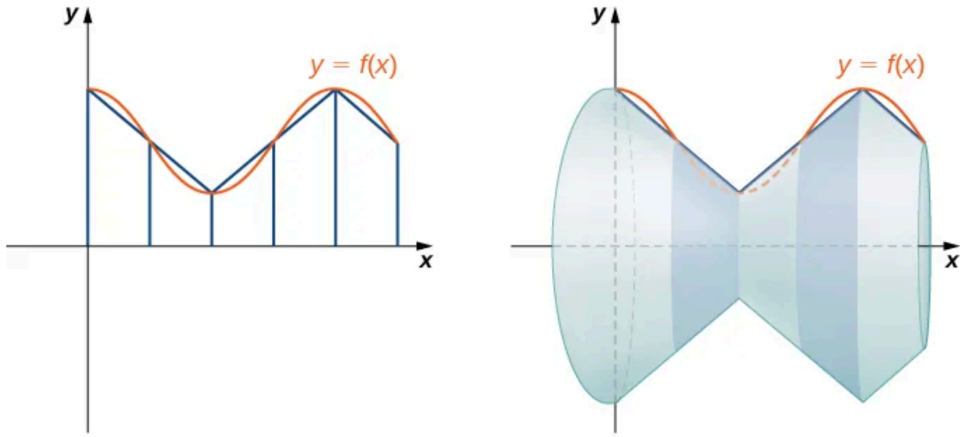
3. Let $f(x) = x^2$. Approximate the arc length of the graph of f over the interval $[1,3]$ with an absolute error of no more than 0.01.

2 Area of a Surface of Revolution

Intuitively, *surface area* is the total area of the outer layer of an object. Here, we use a definite integral to calculate the surface area of the solid of revolution obtained by revolving the graph of a nonnegative function $f(x)$ defined on $[a,b]$ about the x-axis. **Again, we assume that $f'(x)$ exists and is continuous on $[a,b]$.**

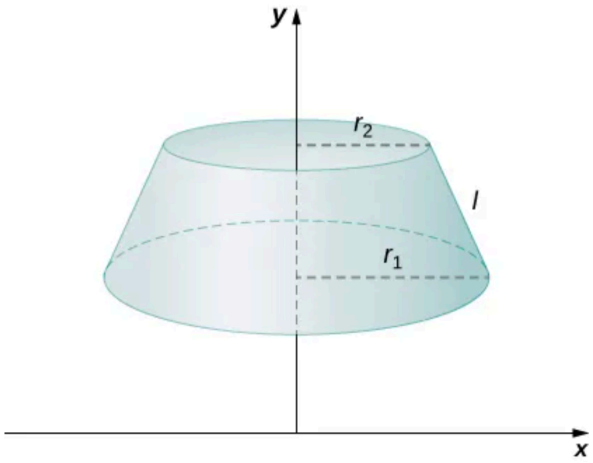


As we did in the previous section, let $P = \{x_i\}$ be a partition of $[a,b]$ into n subintervals of equal length Δx. For each $i = 1,\ldots,n$, construct the line segment from the point $(x_{i-1}, f(x_{i-1}))$ to the point $(x_i, f(x_i))$. Now, revolve these line segments about the x-axis to generate an approximation to the surface of revolution, as shown below:



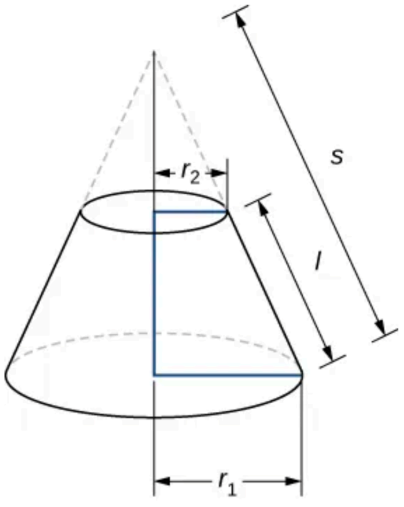
Each segment, when revolved about x-axis, produces a piece of a cone, called a *frustum* of a cone.

To find the lateral surface area of the frustum, let r_1 and r_2 be the radii of the wide end and the narrow end of the frustum, respectively, and let l be the slant height of the frustum, as shown in the figure below:



Recall that the lateral surface area of a cone is $\pi r_1 s$, where r_1 is the radius of the base and s is the slant height of the cone (see the figure below). To find the lateral surface area of the frustum, we subtract from this formula the lateral surface area of the smaller cone:

$$\text{Lateral Surface Area} = \pi r_1 s - \pi r_2 (s - l) \tag{1}$$



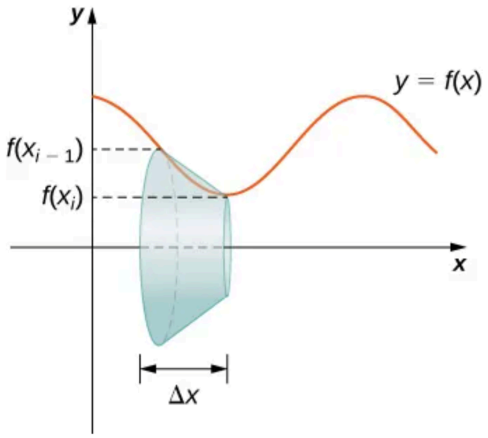
4. Use similar triangles so solve for s in terms of r_1, r_2, and l.

5. Substitute your expression for s into Equation (1) and simplify to show that the lateral surface area of the frustum can be written as

$$\text{Lateral Surface Area} = \pi(r_1 + r_2)l \tag{2}$$

If we are looking at the ith frustum, then $r_1 = f(x_{i-1})$, $r_2 = f(x_i)$ and $l = \sqrt{(\Delta x)^2 + (\Delta y_i)^2} = \Delta x\sqrt{1 + \left(\frac{\Delta y_i}{\Delta x}\right)^2}$ (see the figure below), so Equation (2) becomes

$$S_i = \pi(f(x_{i-1}) + f(x_i))\Delta x\sqrt{1 + \left(\frac{\Delta y_i}{\Delta x}\right)^2}. \tag{3}$$



As before, by the Mean Value Theorem there exists $x_i^* \in [x_{i-1}, x_i]$ such that $f'(x_i^*) = \frac{\Delta y_i}{\Delta x}$, so we can write

$$S_i = \pi(f(x_{i-1}) + f(x_i))\Delta x\sqrt{1 + \left(f'(x_i^*)\right)^2} \tag{4}$$

Since f is continuous, $f(x)$ doesn't change very much over $[x_{i-1}, x_i]$ when Δx is small (which will be the case if n is large enough), so we approximate $f(x_{i-1}) \approx f(x_i) \approx f(x_i^*)$; then Equation (4) becomes

$$S_i = 2\pi f(x_i^*)\Delta x\sqrt{1 + (f'(x_i^*))^2}. \tag{5}$$

The approximate surface area of the entire surface of revolution is then

$$\text{Surface Area} \approx \sum_{i=1}^{n} 2\pi f(x_i^*)\sqrt{1 + (f'(x_i^*))^2}\Delta x,$$

and we define the exact surface area S to be

$$\boxed{S = \lim_{n\to\infty}\sum_{i=1}^{n} 2\pi f(x_i^*)\sqrt{1 + (f'(x_i^*))^2}\Delta x = \int_a^b 2\pi f(x)\sqrt{1 + (f'(x))^2}dx}.$$

6. Let $f(x) = \sqrt{x}$ over the interval $[1,4]$. Find the surface area of the solid of revolution generated by revolving the graph of $f(x)$ about the x-axis.

7. Let $f(x) = y = \sqrt[3]{3x}$. Consider the portion of the curve where $0 \leq y \leq 2$. Find the surface area of the solid of revolution generated by revolving the graph of $f(x)$ about the y-axis.